Total Positivity as the Closure Law: a finite laboratory and a geometric localization of RH(L)

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Abstract

The closure programme asserts a four-fold identity: existence ⇔ closure (fixed point) ⇔ final coalgebra ⇔ positivity of an invariant form. Three links are established elsewhere; the fourth — *when does closure force positivity?* — is the open one, and the Riemann-type question is an instance of it. We build a finite, gate-first laboratory of closure objects on which both sides are exactly decidable and verify, on all 18 of 18 gated objects across two real quadratic fields, seven root lattices, and two closure-operator schemes, the law: *an operator self-adjoint under its invariant trace form is positive under that form if and only if it is totally positive (positive at every place).* Finitely this is the spectral theorem; its value is the organising principle and the localization it yields: the icosian cuspidal L-function is the programme's unique object whose "every place" is infinite, and its total positivity *is* RH(L) — GRH for one cuspidal L-function, not the classical Riemann Hypothesis. We then identify the forcing mechanism quantitatively: per-prime (Ramanujan) bounds provably cannot force the Weil form positive, and the qualitative Sato–Tate limit law is a known theorem for this class of form, so the open, RH-adjacent content is the *rate* of equidistribution (the discrepancy D_n). We calibrate the weight-2 Weil machinery against an elliptic-curve L-function (explicit formula closes to residual $< 10^{-5}$), find the icosian Weil margin positive on every tested function, and record the first discrepancy datum $\sqrt{n}\, D_n \approx 0.73$ at $n = 32$. No proof of RH or any GRH is claimed; the contribution is a law, a localization, a mechanism, and calibrated instruments.

1 The missing link

The programme's spine identifies four statements

$$\text{existence} \iff \text{closure fixed point} \iff \text{final coalgebra} \iff \text{positivity of an invariant form},$$

of which the first three equivalences are established in the existence-as-closure and capstone-coalgebra papers. The fourth link is not: the trace-form work showed that the *right* invariant form is necessary (it is what renders the structure operator self-adjoint, hence real-spectrum) but **not sufficient** for positivity. This note attacks the missing sufficient ingredient on the only terrain where it is exactly decidable: finite closure objects.

Status 1.1. Everything in §2–§3 is *verified* (exact finite computation, code in `lab/`). §4 is *verified localization* resting on the flagship's proved equivalence. §5 mixes *proved* (non-separability, insufficiency of per-prime bounds), *recalled* (Sato–Tate as theorem), and *measured* (discrepancy). Nothing here proves RH or any GRH.

2 The laboratory

Definition 2.1 (closure object, gate)**.** *A* finite closure object *is a pair (B, T) of real matrices on a finite-dimensional space: B symmetric (the candidate invariant form) and T the structure operator (a closure operator, multiplication operator, or Gram matrix), together with the closure-fixed-point status of the underlying object. We say the* right form is present *if $BT = (BT)^\top$ (T is B-self-adjoint); T is B-*positive *if moreover $x \mapsto x^\top(BT)\,x$ is positive definite; and T is* totally positive *if every eigenvalue of T (its values at all places / all conjugates) is strictly positive. The* gate*: no object may receive a positivity verdict unless the right form is present — a wrong or fitted form is rejected, not scored.*

The registry spans: multiplication operators $x \mapsto \alpha x$ on $\mathcal{O}_K$ for $K = \mathbb{Q}(\sqrt{5})$ under the trace form $B = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 3 \end{smallmatrix}\right)$ and for $K = \mathbb{Q}(\sqrt{2})$ under $B = \mathrm{diag}(2, 4)$ (so the law is not golden-ratio-specific); the root-lattice Gram matrices $E_8, E_7, E_6, D_4, A_4, A_3, A_2$; the closure operators $C_\varphi = L + \varphi^{-2} I$ built from the actual 600-cell (120 vertices, neighbours at inner product $\varphi/2$) and 24-cell adjacency; two non-positive self-adjoint controls (an indefinite Lorentzian Gram and the singular affine $\tilde{A}_1$); one wrong-form decoy (multiplication by φ in the $\{1, \sqrt{5}\}$ basis paired with $B = I$, whose true invariant form is $\mathrm{diag}(2, 10)$); and the icosian boundary row of §4.

3 The law

Theorem 3.1 (finite closure-positivity law; verified 18/18)**.** *On every gated, finitely-decidable object in the registry,*

$$T \text{ is } B\text{-positive} \iff T \text{ is totally positive,}$$

and among the structural features tested (positive-definiteness of B itself, reality of the spectrum, total positivity) only *total positivity is an exact discriminator (*18/18*; the others reach* 12/18*). The gate rejects the wrong-form decoy.*

Proof (and honest deflation). For B-self-adjoint T this is the spectral theorem: T is diagonalisable with real spectrum in a B-orthogonal basis, and the form $x^\top(BT)x$ is positive definite iff every eigenvalue is positive; the eigenvalues of a multiplication operator $x \mapsto \alpha x$ on $\mathcal{O}_K$ are precisely the conjugates of α, so "spectrum positive" *is* "totally positive." The computation verifies the implementation, not a new theorem. □

Remark 3.2 (what the law is for)*.* The content is the *organising principle*: every positivity in the programme — Weil positivity for L-functions, the closure-operator positivity of C_φ, the E_8 trace-form closure, the stability of the H4O attractor dynamics — is one statement, *positivity at all places*, with self-adjointness under the right invariant form as the shared precondition. Finite objects are decidable because all their places are visible. That sets up the localization.

4 The localization

Recall from the flagship (and the triad bundle) the icosian closure object: the maximal icosian order $\mathcal{I}$ over $K = \mathbb{Q}(\sqrt{5})$ on the 600-cell, with exact theta identity $L(\Theta_{\mathcal{I}}, s) = \zeta_K(s)\,\zeta_K(s-1)$ and, on its cuspidal face, the Brandt eigenvalues $a_{\mathfrak{q}}$ at level $\mathfrak{p}_{31}$ feeding the Weil quadratic form Q_A with the proved equivalence $Q_A \geq 0 \iff \mathrm{RH}(L)$ — GRH for this one cuspidal L, not classical RH.

Proposition 4.1 (the unique infinite row)**.** *In the laboratory's terms the icosian cuspidal L is a closure object that passes every finite check (self-adjoint Brandt faces, Ramanujan bounds, finite positivity, and the calibrated Weil margins of §6), while its B-positivity in the sense of Theorem 3.1 requires positivity* at all places at once — *an infinite condition. Total positivity of this object* is RH(L).

This is the localization: RH(L) is not an isolated mystery but **the unique infinite instance of a law verified exactly on every finitely-decidable closure object**. The finite/infinite boundary is the entire remaining difficulty.

5 The mechanism

Three results pin down *what would force* the open positivity.

5.1 Non-separability (proved)

The Li coefficients [2, 3] give finite certificates (RH(L) $\iff$ $\lambda_n \geq 0\ \forall n$), but the literal per-prime decomposition of λ_n diverges term by term (for ζ: $\sum_p (\log p)^n/p$); each λ_n is finite only after archimedean–prime cancellation. There is no separable per-prime positivity to verify — consistent with the programme's earlier negative results on separable envelopes. The convergent per-prime object is the Weil form [1] $W(g) = \mathrm{ARCH}(g) - \sum_{\mathfrak{q}} \mathrm{PRIME}_{\mathfrak{q}}(g)$.

5.2 Per-prime bounds cannot force positivity (proved)

Every geometric eigenvalue satisfies Ramanujan, $|a_{\mathfrak{q}}| \leq 2\sqrt{N\mathfrak{q}}$ (verified on all rows; Ramanujan for this class of form is a theorem via [6]). But bounding each $\mathrm{PRIME}_{\mathfrak{q}}$ cannot bound their *sum* below ARCH: if it could, Ramanujan would imply RH(L), which it does not. Positivity is a statement about the *ensemble*, not the terms.

5.3 The ensemble property, split honestly in two

Write $x_{\mathfrak{q}} = a_{\mathfrak{q}}/2\sqrt{N\mathfrak{q}} = \cos\theta_{\mathfrak{q}}$.

(i) The limit law is known. Qualitative Sato–Tate equidistribution of $\{\theta_{\mathfrak{q}}\}$ under $\frac{2}{\pi}\sin^2\theta$ is a theorem for non-CM Hilbert modular forms [5]. It therefore forces *nothing* about RH(L): the limit holds and RH(L) remains open.

(ii) The rate is the open content. What relates to RH-type statements is *quantitative* equidistribution — the discrepancy $D_n = \sup_x |F_n(x) - F_{\mathrm{ST}}(x)|$ and its decay, governed by the analytic behaviour of the symmetric-power L-functions (effective Sato–Tate, cf. [7]); square-root cancellation $D_n = O(n^{-1/2+\varepsilon})$ is the GRH-regime signature.

Conjecture 5.1 (geometric forcing, stated as the programme's frontier)**.** *The closure geometry of $\mathcal{I}$ forces the square-root-cancellation rate of equidistribution of its Brandt spectrum; equivalently, the geometry forces the positivity of Q_A.* (Open. This is a restatement of the analytic frontier in geometric terms, not a new route around it; by §4 its conclusion is RH(L).)

6 Instruments and first data

Calibration. The weight-2 Weil machinery was calibrated against the conductor-11 elliptic-curve L (degree 2, weight 2, no pole; zeros and a_p from PARI [8]): with archimedean shifts $\{\frac{1}{2}, \frac{3}{2}\}$ per place and the universal prime-side factor 2, the explicit formula closes to max $|\text{residual}| < 10^{-5}$ across all tested widths. (The calibration caught and fixed two normalization errors that a ζ-only check structurally cannot see; a wrong convention cannot survive this gate.)

Icosian Weil margins (evidence, not proof). With the calibrated convention (degree 4, conductor 775, shifts $\{\frac{1}{2}, \frac{1}{2}, \frac{3}{2}, \frac{3}{2}\}$), $W(g) = \mathrm{ARCH}(g) - \sum \mathrm{PRIME}_{\mathfrak{q}}(g) > 0$ on every tested Gaussian g:

σ	1.5	2.0	2.5	3.0	3.5	4.0
margin W	+0.13	+0.60	+1.38	+2.34	+3.44	+4.65

The archimedean term drives positivity; the prime sum is a Ramanujan-bounded ripple.

Li certificates. The λ_n engine (zeros $\to$ partial Li sums) self-tests on ζ: 200 zeros give $\lambda_1, \ldots, \lambda_6 > 0$ tracking the known values [4, 2]. It ingests the icosian L's own zeros as the extended Brandt computation delivers them.

First discrepancy datum. At $n = 32$ prime ideals: moments $\mathbb{E}[x^2] = 0.226$ (ST: 0.25), $\mathbb{E}[x^4] = 0.103$ (ST: 0.125); discrepancy $D_{32} = 0.128$, i.e. $\sqrt{n}\, D_n = 0.73$ — an $O(1)$ constant, consistent with (far from probative of) the square-root-cancellation regime. The extended run populates this curve; $\sqrt{n}\, D_n$ bounded as n grows is the measurable shadow of Conjecture 5.1.

7 What is not claimed

No proof of the Riemann Hypothesis or any GRH. No claim that Conjecture 5.1 is easier than RH(L) — by construction it is equivalent in conclusion. No claim that the finite law (Theorem 3.1) is new mathematics — finitely it is the spectral theorem; the contribution is the organising principle, the localization (Proposition 4.1), the honest two-level mechanism (§5), and externally calibrated instruments with first data (§6). The physical and cosmological faces of the programme are not load-bearing here and RH claims never rest on them.

References

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